# Multi-period planning

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Lecture 4

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## Contents

# 1 Planning problems

- Planning future activities
- Decide the values of planning variables, like workforce, capacities, production etc. for the future
- The future is divided into a number of discrete time periods, within a given time planning horizon

## 1.1 Link between periods

Normally, the time periods (days, weeks, months) are linked in some way:

- in production / supply chain planning:

$$\text{inventory}(t) = \text{inventory}(t-1) + \text{production}(t) - \text{demand}(t)$$

- in manpower planning:

$$\text{workforce}(t) = \text{workforce}(t-1) + \text{people hired}(t) - \text{people fired}(t)$$

- In financial planning:

$$\text{cash balance}(t) = \text{cash balance}(t-1) + \text{incoming cash flow}(t) - \text{payments}(t)$$

## 2 Example

- 1 product
- 3 periods

| | period 1 | period 2 | period 3 |
|---|---|---|---|
| demand | 4 | 9 | 14 |
| initial inventory | 3 | 0 | 0 |
| production capacity | 10 | 10 | 10 |
| production cost | 4 | 7 | 8 |
| inventory cost | 2 | 2 | 2 |

- Optimization problem:
  - Minimize total costs = sum production costs + sum inventory costs
  - s.t. production must be less than or equal to production capacity in each period, inventory balance constraint, production and inventory must be non-negative in each period
- variables:
  - $X_t$: production quantity in period $t$
  - $I_t$: inventory by the end of period $t$
- Parameters:
  -
  - $c_t$: production cost in period $t$
  - $i_t$: inventory cost in period $t$
  - $\text{cap}_t$: production capacity for period $t$
  - $d_t$: demand for period $t$
  - init: initial inventory
- model

$$
\begin{gathered}
\min \sum_t c_t X_t + i_t I_t \\
\text{s.t.} \\
X_t \leq \text{cap}_t \quad \forall t \\
I_t = I_{t-1} + X_t - d_t \quad \forall t \geq 2 \\
I_1 = \text{init} + X_1 - d_1 \\
X_t, I_t \geq 0 \quad \forall t
\end{gathered}
$$

## 3 Example extension: multiple products

Parameters:

- $c_{p,t}$: production cost for product $p$ in time period $t$
- $i_{p,t}$: inventory holding cost for product $p$ in time period $t$
- $\text{cap}_t$: production capacity for period $t$
- $d_{p,t}$: demand for product $p$ in time period $t$
- $\text{init}_p$: initial inventory for product $p$

Variables:

- $X_{p,t}$: production quantity for product $p$ in period $t$
- $I_{p,t}$: inventory for product $p$ in period $t$

Model:

$$\min \sum_p \sum_t c_{p,t} X_{p,t} + i_{p,t} I_{p,t}$$

$$\text{s.t.}$$

$$\sum_p X_{p,t} \leq \text{cap}_t \quad \forall t$$

$$I_{p,t} = I_{p,t-1} + X_{p,t} - d_{p,t} \quad \forall p \quad \forall t \geq 2$$

$$I_{p,1} = \text{init}_p + X_{p,1} - d_{p,1} \quad \forall p$$

$$X_{p,t}, I_{p,t} \geq 0 \quad \forall p, t$$

# 4 Basic production planning, one customer

Parameters:

- $pc_t$: production cost per unit produced in period $t$
- $sc_t$: storage cost for storing per unit in inventory by the end of period $t$
- $\text{cap}_t$: production capacity in period $t$
- $d_t$: demand in period $t$

Variables:

- $X_t$: production quantity in period $t$
- $I_t$: inventory by the end of period $t$

Model:

$$\min \sum_t pc_t X_t + sc_t I_t$$

$$\text{s.t.}$$

$$X_t \leq \text{cap}_t \quad \forall t$$

$$I_t = I_{t-1} + X_t - d_t \quad \forall t \geq 2$$

$$I_1 = X_1 - d_1$$

$$X_t, I_t \geq 0 \quad \forall t$$

# 5 Basic production planning, one factory

Parameters:

- $pc_t$: production cost per unit produced in period $t$
- $sc_t$: storage cost for storing per unit in inventory by the end of period $t$
- $\text{cap}_t$: production capacity in period $t$
- $d_{k,t}$: demand for customer $k$ in period $t$

Variables:

- $X_t$: production quantity in period $t$
- $I_t$: inventory by the end of period $t$

Model:

$$\min \sum_t pc_t X_t + sc_t I_t$$

$$\text{s.t.}$$

$$X_t \leq \text{cap}_t \quad \forall t$$

$$I_t = I_{t-1} + X_t - \sum_k d_{k,t} \quad \forall t$$

$$X_t, I_t \geq 0 \quad \forall t$$

# 6 Production planning, coordination of multiple factories $i$

Parameters:

- $pc_{i,t}$: production cost per unit produced in period $t$
- $sc_{i,t}$: storage cost for storing per unit in inventory at $i$ by the end of period $t$
- $t_c i, k, t$: transportation cost per unit from $i$ to $k$ in period $t$
- $\text{cap}_{i,t}$: production capacity at factory $i$ in period $t$
- $d_{k,t}$: demand for customer $k$ in period $t$

Variables:

- $X_{i,t}$: production quantity at factory $i$ in period $t$
- $I_{i,t}$: inventory at factory $i$ in period $t$
- $Z_{i,k,t}$: transported quantity from $i$ to $k$ in period $t$

Model:

$$\min \sum_i \sum_t pc_{i,t} X_{i,t} + sc_{i,t} I_{i,t} + \sum_i \sum_t \sum_k tc_{i,k,t} Z_{i,k,t}$$

$$\text{s.t.}$$

$$X_{i,t} \leq \text{cap}_{i,t} \quad \forall i, t$$

$$I_{i,t} = I_{i,t-1} + X_{i,t} - \sum_k Z_{i,k,t} \quad \forall i, t$$

$$\sum_i Z_{i,k,t} = d_{k,t} \quad \forall k, t$$

# 7 Economies of scale

- Economies of scale of the order of quantity is a very common phenomenom
- In a production process, there is normally some loss of efficiency when switching between products, i.e., we would like to maximize the lot size when we produce a given product.
- Also in transportation, there is normally economies of scale. Driving a dairy truck a given distance has the same cost regardless of how many bottles of milk there are in the back. That is, a full truck will have a lower cost per bottle
- In purchasing, the economies of scale is often represented by a given cost